EE4-07 Coding Theory

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. (Finite Fields)

(a) Let $a(x) = (x^2+1)\,b(x) \mod x^4+x+1$, where all the involved polynomials are in $\mathbb{F}_2[x]$. Suppose that $\deg(a(x)) = 0$. Find $a(x)$ and $b(x)$ by using Euclidean algorithm and Bézout's identity. [5]

(b) Let $\mathcal{C} \subset \mathbb{F}_2^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_2^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_2^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over an binary symmetric channel. Let the received word be $\boldsymbol{y}_1 = [0\ 0\ 1\ 1\ 1\ 1\ 0]$. Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$, and the corresponding transmitted message $\hat{\boldsymbol{m}}_1$. [3]

iv Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over an binary erasure channel. Let the received word be $\boldsymbol{y}_2 = [?\ 1\ 1\ 0\ ?\ 0\ 1]$. Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$ and the message $\boldsymbol{m}_2$. [3]

(c) Let $\mathcal{C} \subset \mathbb{F}_3^7$ be a linear code. (Note that $\mathbb{F}_3$ is used rather than $\mathbb{F}_2$.) Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_3^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_3^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

(d)

i Find whether $x^2+1 \in \mathbb{F}_2[x]$ is irreducible or not. Prove your answer. [2]

ii Find whether $x^2 + 1 \in \mathbb{F}_3[x]$ is irreducible or not. Prove your answer. [2]

(e) Prove that in $\mathbb{F}_p[x]$, the equality that $(x+1)^p = x^p + 1$ holds.

Remark: You are allowed to use Euclid's Lemma: if integers a, b, c satisfy the properties that $a|(b \cdot c)$ and $\gcd(a, b) = 1$, then $a|c$. [2]

2. (Reed-Solomon codes)

Remark: You are allowed to use the facts that the distance of a linear code is the minimum number of the columns of its parity matrix $\boldsymbol{H}$ that are linearly dependent, and that the distance of a linear code is also the minimum number of nonzero elements in nonzero codewords.

(a) State and prove the Singleton bound for linear codes $\mathcal{C}[n,k,d]$. Codes that attain the singleton bound are maximum distance separable (MDS). [3]

(b) Define Reed-Solomon codes in the following way. Let $\mathbb{F}_q$ be a finite field and α be a primitive element of $\mathbb{F}_q$. Let $n = q-1$. For any polynomial $f(x) \in \mathbb{F}_q[x]$, define the evaluation map $\text{eval}(f(x))$ by mapping $f(x)$ to a vector $\boldsymbol{c} \in \mathbb{F}_q^{n-1}$:

$$f(x) \mapsto \boldsymbol{c} = [c_1, \cdots, c_n], \text{ where } c_i = f\left(\alpha^{i-1}\right).$$

An $[n,k]$ Reed-Solomon code is defined as

$$\mathcal{C}[n,k] = \{\text{eval}(f(x)),\ 0 \leq \deg f(x) \leq k-1\}.$$

i Prove that Reed-Solomon codes are linear codes. [3]

ii Prove that Reed-Solomon codes attain the Singleton bound. That is, Reed-Solomon codes are MDS. [3]

iii Based on the definition, find the corresponding generator matrix $\boldsymbol{G}$. [3]

iv The parity check matrix of a Reed-Solomon code can be written as

$$\boldsymbol{H} = \begin{bmatrix} 1 & \alpha & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{n-k} & \cdots & \alpha^{(n-k)(n-1)} \end{bmatrix},$$

where the (j,i)-th entry is given by $h_{j,i} = \alpha^{(i-1)j}$, $1 \leq j \leq n-k$ and $1 \leq i \leq n$. Let $\boldsymbol{y} = \boldsymbol{c} + \boldsymbol{e}$ be the received word, where $\boldsymbol{c}$ is the transmitted codeword from a Reed-Solomon code and $\boldsymbol{e} = [e_1, \cdots, e_n]$ is the error vector introduced by the channel. Define the index set of errors by $\mathcal{I} = \{i :\ e_i \neq 0,\ 1 \leq i \leq n\}$. Define the syndrome vector $\boldsymbol{s} = \boldsymbol{y}\boldsymbol{H}^T = [s_1, \cdots, s_{n-k}]$. Give the general expression for s_j, $1 \leq j \leq n$, using the error symbols e_i, $i \in \mathcal{I}$. [2]

v Define the syndrome polynomial $S(z) = \sum_{j=0}^{n-k-1} s_{j+1} z^j$. Prove that

$$S(z) \equiv \sum_{i \in \mathcal{I}} \frac{e_i \alpha^{i-1}}{1 - \alpha^{i-1} z} \bmod z^{n-k}.$$

[5]

vi Define the error locator polynomial

$$L(z) = \prod_{i \in \mathcal{I}} \left(1 - \alpha^{i-1} z\right),$$

and the error evaluator polynomial

$$E(z) = \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \prod_{j \in \mathcal{I} \setminus \{i\}} \left(1 - \alpha^{j-1} z\right).$$

Specify the key equation used for Reed-Solomon decoding. [2]

vii Describe the process to use the syndrome polynomial $S(z)$ to find both the error locator polynomial $L(z)$ and the error evaluator polynomial $E(z)$. [4]

3. (Cryptography)

Remark: You are allowed to use the fact that for all nonzero elements $\alpha, \beta \in \mathbb{F}_q$, their product $\alpha \cdot \beta$ is also nonzero.

(a) Fermat's Little Theorem states that for every $\beta \in \mathbb{F}_q^*$, one has $\beta^{q-1} = 1$. It can be proved using the following arguments.

i For every $\beta \in \mathbb{F}_q^*$, define $\beta\mathbb{F}_q^* = \{\beta\beta_1, \cdots, \beta\beta_{q-1}\}$. Prove that $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$. [2]

ii Prove that $\beta^{q-1} = 1$. [2]

(b) Prove that for every $\beta \in \mathbb{F}_q^*$, it holds that $\operatorname{ord}(\beta) \mid (q-1)$. [3]

(c) Consider the prime number $p = 311$.

i Find the missing values in the table below.

x	1	2	5	10	31	62	155
$2^x \bmod 311$	2	4	32	91	?	?	?
$17^x \bmod 311$	17	289	142	260	?	?	?

[6]

ii Between the numbers 2 and 17, which one is a primitive element in $\mathbb{F}_{311}$? Explain the reason. [4]

iii Find $\log_{17} 260 \bmod 311$ and $\log_{17} 36 \bmod 311$. [2]

(d) Consider RSA public key cryptography. Let the two involved prime numbers be $p = 311$ and $q = 313$.

i Let $(n, e) = (97343, 23)$ be the public key. Define $t = (p-1)(q-1)$. The private key d satisfies $e \cdot d = 1 \bmod t$. Find the corresponding private key d. [3]

ii Suppose that Alice would like to encrypt the message $m = 11$ using the public key (n, e). Find the encrypted word c. [3]

Solution of Question 1.

(a) Apply the Euclidean algorithm to the polynomials $x^4 + x + 1$ and $x^2 + 1$. One has

$$\begin{aligned} x^4 + x + 1 &= \left(x^2 + 1\right) \cdot \left(x^2 + 1\right) + x \\ x^2 + 1 &= x \cdot x + 1. \end{aligned}$$

As a consequence, [3/5]

$$\begin{aligned} 1 &= \left(x^2 + 1\right) + x \cdot x \\ &= \left(x^2 + 1\right) + x \cdot \left(\left(x^4 + x + 1\right) + \left(x^2 + 1\right)\left(x^2 + 1\right)\right) \\ &= x\left(x^4 + x + 1\right) + \left(x^3 + x + 1\right)\left(x^2 + 1\right). \end{aligned}$$

Therefore,

$$\begin{aligned} a(x) &= 1, \\ b(x) &= x^3 + x + 1. \end{aligned}$$

[2/5]

(b)

i

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

[2]

ii

$$\boldsymbol{G} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

iii The syndrome vector is given by

$$\boldsymbol{s}_1 = \boldsymbol{y}_1 \boldsymbol{H}^T = [0\ 1\ 1],$$

hence

$$\hat{\boldsymbol{c}}_1 = [0\ 0\ 1\ 1\ 1\ 0\ 0],$$

and

$$\hat{\boldsymbol{m}}_1 = [1\ 1\ 0\ 0]\,.$$

[3]

iv The syndrome vector is given by

$$[0\ 1\ 1\ 0\ 0\ 0\ 1]\,\boldsymbol{H}^T = [1\ 1\ 0]\,.$$

Let c_0 and c_4 be the missing symbols in $\boldsymbol{c}$. Then one has

$$[c_0\ c_4]\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = [1\ 1\ 0]\,.$$

It is clear that $[c_0\ c_4] = [0\ 1]$. Hence $\boldsymbol{c}_2 = [0\ 1\ 1\ 0\ 1\ 0\ 1]$ and $\boldsymbol{m}_2 = [0\ 1\ 0\ 1]$. [3]

(c)

i Here note that $0 - 1 = 2 \in \mathbb{F}_3$. One has

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

[2]

ii From the fact that if $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}]$ then $\boldsymbol{G} = \left[-\boldsymbol{A}^T\ \boldsymbol{I}\right]$. One has

$$\boldsymbol{G} = \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

(d)

i Note that $x^2 + 1 = (x+1)(x+1)$ in $\mathbb{F}_2[x]$. Hence $x^2 + 1 \in \mathbb{F}_2[x]$ is not irreducible. [2]

ii Note that $x^2 + 1 = (x+1)(x+2) + 2$ in $\mathbb{F}_3[x]$. There is no polynomial with degree one that divides $x^2 + 1$ in $\mathbb{F}_3[x]$. $x^2 + 1 \in \mathbb{F}_3[x]$ is irreducible. [2]

(e)

$$(x+1)^p = \sum_{i=0}^{p} \binom{p}{i} x^i.$$

It is clear that $\binom{p}{0} = \binom{p}{p} = 1$. It suffices to prove that $p|\binom{p}{i}$ for all $1 \leq i \leq p-1$. Note that

$$\binom{p}{i} = \frac{p\,(p-1)\cdots(p-i+1)}{i!} \in \mathbb{Z}.$$

That is

$$(i!)\,|p \cdot ((p-1)\cdots(p-i+1))\,.$$

Because $\gcd(i!, p) = 1$, by Euclid's Lemma one has $(i!) \mid (p-1)\cdots(p-i+1)$. Hence $p|\binom{p}{i}$ for all $1 \leq i \leq p-1$. [2]

Solutions of Question 2.

(a) The Singleton bound for linear codes $\mathcal{C}[n,k,d]$ states that

$$d \leq n-k+1.$$

To prove it, consider the corresponding parity check matrix $\boldsymbol{H} \in \mathbb{F}_q^{(n-k)\times n}$. [1/3]
The code distance d is defined as the minimum number of linearly dependent columns of $\boldsymbol{H}$. On the other hand, for the parity check matrix $\boldsymbol{H} \in \mathbb{F}_q^{(n-k)\times n}$, every $n-k+1$ columns have to be linearly dependent. Therefore,

$$d \leq n-k+1.$$

[2/3]

(b)

i Let $\boldsymbol{c}_1 = \text{eval}(f_1)$ and $\boldsymbol{c}_2 = \text{eval}(f_2)$ where $\deg f_1 \leq k-1$ and $\deg f_2 \leq k-1$. Then $\alpha\boldsymbol{c}_1+\beta\boldsymbol{c}_2 = \text{eval}(g)$ with $g = \alpha f_1+\beta f_2$. Since $\deg g \leq k-1$, $\text{eval}(g) \in \mathcal{C}$. [3]

ii A polynomial of degree $\leq k-1$ can have at most $k-1$ zeros. Hence, $\forall \boldsymbol{c} \in \mathcal{C}$ s.t. $\boldsymbol{c} \neq 0$, $\boldsymbol{c} = \text{eval}(f)$ has weight at least $n-k+1$. [3]

iii Based on the definition, the generator matrix of a Reed-Solomon code is given by

$$\boldsymbol{G} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \alpha & \cdots & \alpha^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{k-1} & \cdots & \alpha^{(k-1)(n-1)} \end{bmatrix}.$$

[3]

iv It is clear that $\boldsymbol{s} = \boldsymbol{y}\boldsymbol{H}^T = (\boldsymbol{c}+\boldsymbol{e})\boldsymbol{H}^T = \boldsymbol{e}\boldsymbol{H}^T$. Hence

$$s_j = \sum_{i=1}^{n} e_i h_{j,i} = \sum_{i\in\mathcal{I}} e_i \alpha^{(i-1)j}.$$

[2]

v By definition,

$$\begin{aligned} S(z) &= \sum_{j=0}^{n-k-1} s_{j+1} z^j \\ &= \sum_{j=0}^{n-k-1} \left(\sum_{i \in \mathcal{I}} e_i \alpha^{(i-1)(j+1)} \right) z^j \\ &= \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \left(\sum_{j=0}^{n-k-1} \left(\alpha^{i-1} z \right)^j \right) \\ &\equiv \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \left(\sum_{j=0}^{\infty} \left(\alpha^{i-1} z \right)^j \right) \mod z^{n-k} \\ &= \sum_{i \in \mathcal{I}} \frac{e_i \alpha^{i-1}}{1 - \alpha^{i-1} z}. \end{aligned}$$

[5]

vi The key equation used for Reed-Solomon decoding is given by

$$E(z) = L(z) \cdot S(z) \mod z^{n-k}.$$

[2]

vii Apply Euclidean algorithm to the polynomials z^{n-k} and $S(z)$ until the the degree of the remainder polynomial is at most $\left\lfloor \frac{n-k}{2} \right\rfloor - 1$. [1/4]
According to Bézout's identity, one has

$$a(z) = b(z) S(z) + c(z) z^{n-k},$$

for some polynomials $a(z)$, $b(z)$, and $c(z)$. [2/4]
Denote the constant term of $b(z)$ by d. Then $L(z) = d^{-1} b(z)$ and $E(z) = d^{-1} a(z)$. [1/4]

Solutions of Question 3.

(a)

i As all elements in $\beta\mathbb{F}_q^*$ has the form $\beta\beta_i \neq 0$, it is clear that $\beta\mathbb{F}_q^* \subseteq \mathbb{F}_q^*$. For any given $\beta_i \neq \beta_j$, $\beta(\beta_i - \beta_j) \neq 0$ or equivalently $\beta\beta_i \neq \beta\beta_j$. Therefore, $\left|\beta\mathbb{F}_q^*\right| = q-1$. The only possibility is therefore $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$. [2]

ii Since $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$, it holds that

$$\prod_{\gamma\in\beta\mathbb{F}_q^*} \gamma = \prod_{\gamma\in\mathbb{F}_q^*} \gamma.$$

That is

$$\beta^{q-1} \prod_{\gamma\in\mathbb{F}_q^*} \gamma = \prod_{\gamma\in\mathbb{F}_q^*} \gamma.$$

Note that $\prod_{\gamma\in\mathbb{F}_q^*}\gamma$ is a nonzero element and its multiplicative inverse exists. Hence, $\beta^{q-1} = 1$. [2]

(b) Let $t = \mathrm{ord}(\beta)$. Suppose that $t \nmid (q-1)$. Then there exists $0 < r < t$ such that $(q-1) = \ell \cdot t + r$ for an integer ℓ. By Fermat's Little Theorem, $1 = \beta^{q-1} = \beta^{\ell\cdot t+r} = \beta^r$. This contradicts the definition that $t = \mathrm{ord}(\beta)$ is the smallest integer such that $\beta^t = 1$. Hence $t|(q-1)$. [3]

(c)

i The complete table is given by

x	1	2	5	10	31	62	155
$2^x \bmod 311$	2	4	32	91	**36**	**52**	**1**
$17^x \bmod 311$	17	289	142	260	**305**	**36**	**310**

The detailed computations are given as follows.

$\mathrm{mod}(2^{31}, 311) = \mathrm{mod}\left((2^{10})^3 \cdot 2, 311\right) = \mathrm{mod}(91^3 \cdot 2, 311) = 36.$

$\mathrm{mod}(2^{62}, 311) = \mathrm{mod}\left((2^{31})^2, 311\right) = \mathrm{mod}(36^2, 311) = 52.$

$\mathrm{mod}(2^{155}, 311) = \mathrm{mod}\left((2^{31})^5, 311\right) = \mathrm{mod}(36^5, 311) = 1.$

$\mathrm{mod}(17^{31}, 311) = \mathrm{mod}\left((17^{10})^3 \cdot 17, 311\right) = \mathrm{mod}(260^3 \cdot 17, 311) = 305.$

$\mathrm{mod}(17^{62}, 311) = \mathrm{mod}\left((17^{31})^2, 311\right) = \mathrm{mod}(305^2, 311) = 36.$

$\mathrm{mod}(17^{62}, 311) = \mathrm{mod}\left((17^{31})^5, 311\right) = \mathrm{mod}(305^5, 311) = 310.$ [6]

Remark. In order to compute for example $\mod(17^{62}, 311)$, one can use the fact that $17^{62} = (17^{31})^2 = (17^{10})^3 \times 17^2$. Clearly a direct computation 17^{62} may give wrong answers due to the finite many bits used to present a number in a calculator.

ii The number 17 is a primitive element in $\mathbb{F}_{311}$.
The number 2 is clearly not a primitive element in $\mathbb{F}_{311}$ as $2^{155} = 1$. [2/4]
The number 17 is a primitive element because $\mathrm{ord}(17) = 310$. To check that, note that $310 = 2 \times 5 \times 31$. The only possible values for $\mathrm{ord}(17)$ are $1, 2, 5, 10, 31, 62, 155, 310$. From the third line of the above table, it is clear that $\mathrm{ord}(17) = 310$. [2/4]

iii From the table, it is clear that

$$10 = \log_{17} 260 \bmod 311,$$

and

$$62 = \log_{17} 36 \bmod 311.$$

[2]

(d)

i It is clear that $t = 310 \times 312 = 96720$. Apply Euclidean algorithm. One has

$$\begin{aligned} 96720 &= 23 \times 4205 + 5 \\ 23 &= 5 \times 4 + 3 \\ 5 &= 3 + 2 \\ 3 &= 2 + 1. \end{aligned}$$

Based on Bézout's identity, one has

$$\begin{aligned} 1 &= 3 - 2 = 3 - (5 - 3) = 3 \cdot 2 - 5 \\ &= (23 - 5 \cdot 4) \cdot 2 - 5 = 23 \cdot 2 - 5 \cdot 9 \\ &= 23 \cdot 2 - (96720 - 23 \cdot 4205) \cdot 9 \\ &= -96720 \cdot 9 + 23 \cdot 37847. \end{aligned}$$

Hence

$$d = 37847.$$

[3]

ii

$$c = 11^{23} \bmod 97343.$$

Note that $23 = 16 + 4 + 2 + 1$. Furthermore,

x	1	2	4	8	16
$11^x \bmod 97343$	11	121	14641	9595	74890

$11^{20} = 11^{16} \cdot 11^4 = 90281 \bmod 97343$, and $11^{23} = 90281 \cdot 121 \cdot 11 = 42749 \bmod 97343$. Therefore,

$$c = 42749.$$

[3]